Working Reference Document: Agent-Based Monte Carlo Model of Cancer Growth

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1 Purpose of This Document

This document defines the fixed conceptual, methodological, and bibliographic scope of a small computational project studying the influence of oxygen availability on the growth of cancer and normal cells.

The document is intended to be used as a persistent reference during model development and discussion. It defines what modeling approaches are allowed, what level of biological detail is assumed, and which sources are considered relevant.

This project is exploratory and educational in nature.

2 Methodological Scope

The project is restricted to the following modeling choices:

- Agent-based modeling (ABM) at the cellular level
- Stochastic dynamics implemented via Monte Carlo (MC) sampling
- Qualitative and semi-quantitative analysis of simulation outcomes

The following approaches are explicitly outside the scope of the project unless stated otherwise:

- Fully deterministic continuum models
- Data-driven or machine-learning-based models
- High-fidelity biological or clinical prediction

Monte Carlo sampling is treated as a core modeling tool rather than an auxiliary technique.

3 General Biological Context

Tumor growth is strongly influenced by oxygen availability in tissue. Due to abnormal vasculature and rapid proliferation, tumors often develop hypoxic regions.

Cancer cells typically exhibit:

- Increased tolerance to low oxygen conditions
- Altered proliferation and survival behavior under hypoxia

Normal cells are generally more sensitive to oxygen deprivation. These qualitative differences motivate modeling oxygen-dependent stochastic cell behavior.

No detailed molecular or metabolic pathways are modeled.

4 Conceptual Modeling Framework

The system is modeled at the cellular scale using an agent-based framework:

- Each cell is represented as an individual agent
- Cells exist in a simplified spatial environment
- Each cell undergoes discrete-time updates

At each update step, cell fate decisions (e.g. division, death, quiescence) are determined via Monte Carlo sampling of predefined probabilities.

The model aims to capture population-level trends emerging from stochastic single-cell behavior.

5 Role of Oxygen

Oxygen is treated as an external environmental variable influencing cell behavior.

Key assumptions:

- Oxygen availability affects probabilities of division and death
- Cancer cells retain higher survival and division probabilities under hypoxia
- Normal cells exhibit stronger suppression under low oxygen

Oxygen may be represented in a simplified manner, such as:

- A fixed global parameter
- A spatially varying but static field
- A small number of discrete oxygen states

The representation is chosen for conceptual clarity rather than biological completeness.

6 Parameters and Use of Biological Information

Model parameters are treated as effective quantities inspired by biological observations.

Parameter choices are guided by:

- Experimentally reported ranges
- Relative differences between cancer and normal cells
- Qualitative trends described in review literature

Exact numerical accuracy is not required. Parameters are varied systematically to study trends and robustness of outcomes.

Whenever possible, parameter choices are supported by citations.

7 Relevant Literature and Resources

The following sources are considered central references for biological motivation and modeling inspiration. They are not intended to be reproduced or fully implemented.

- **Grimes et al.**, "The role of oxygen in avascular tumor growth", *PLoS Computational Biology*, 2020. Article link. Associated data and code are available via GitHub as part of a comprehensive modeling framework.
- **Kchakidze et al.**, "Hybrid agent-based modeling of tumor growth", *Computational Biology*, 2020. PubMed link.

These works inform:

- Reasonable abstraction levels for tumor ABMs
- Typical treatment of oxygen effects
- Accepted simplifications in exploratory studies

8 Software Scope

Although some referenced works provide full simulation platforms (e.g. CHESS), this project intentionally avoids reproducing or extending existing software.

Instead:

- A lightweight custom implementation is developed
- Julia is used for implementation
- Only core mechanisms relevant to oxygen-driven growth are included

The focus is on transparency and controllability rather than completeness.

9 Limitations

The model neglects many biological processes, including but not limited to:

- Angiogenesis
- Immune response
- Mechanical interactions
- Detailed metabolism

Results are interpreted qualitatively and comparatively, not predictively.

10 Intended Use

This document serves as:

- A fixed reference for project development
- A constraint on modeling choices
- A shared context for AI-assisted reasoning

All modeling discussions are assumed to operate within the scope defined here unless explicitly stated otherwise.